Chapter 7: Work, Energy, and Energy Resources

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 7

Part I

Work

Outline of Part I: Work

Definition of Work

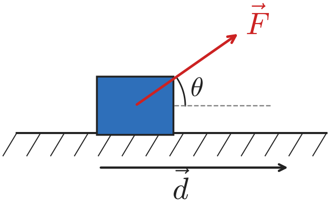
What Is Work?

Work is done on an object when a force acts on it *along* the direction of displacement.

$$W = \vec{F} \cdot \vec{d} = |\vec{F}|\,|\vec{d}| \cos\theta$$

- θ = angle between $\vec{F}$ and $\vec{d}$
- SI unit: **joule** ($\mathrm{J} = \mathrm{N} \cdot \mathrm{m}$)
- W is a *scalar* (positive, negative, or zero)

$$W = Fd\cos\theta$$



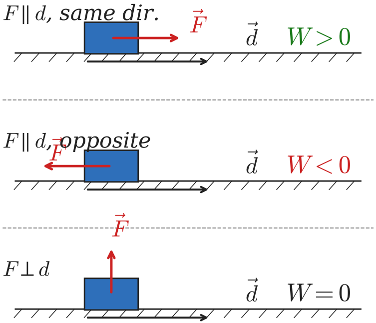
Positive, Negative, and Zero Work

Zero Work

If $\vec{F} \perp \vec{d}$, then $\cos 90° = 0 \;\Rightarrow\; W = 0$.

Examples: normal force on a sliding block; centripetal force on circular orbit; carrying a bag horizontally.

Scenario	θ	**Sign of** W
$\vec{F}$ & $\vec{d}$ same direction	$0°$	$+$
$\vec{F}$ & $\vec{d}$ opposite	$180°$	$-$
$\vec{F}$ perpendicular to $\vec{d}$	$90°$	0



Work: Worked Examples

Bob pushes a lawnmower with $F = 100\,\mathrm{N}$ at $\theta = 37.0°$ below horizontal over $d = 2.50\,\mathrm{m}$. How much work does Bob do?

$$
\begin{aligned}
W &= Fd\cos\theta \\
&= (100)(2.50)\cos(37.0°) \\
\Aboxed{W &= 199\,\mathrm{J}}
\end{aligned}
$$

Check Your Understanding

In which of the following situations is **zero** net work done on the object by the listed force?

(A) A horse pulls a cart forward along a flat road.

(B) A waiter carries a tray horizontally across the room at constant velocity.

(C) The Moon orbits Earth (gravity is the force).

(D) A ball rolls along a frictionless floor (normal force).

Come to lecture to see the answer.

Check Your Understanding

A 50.0 kg rock climber ascends 5.00 m vertically.

How much work does **gravity** do on the climber?

Come to lecture to see the answer.

Part II

Kinetic Energy & the Work-Energy Theorem

Outline of Part II: Kinetic Energy & the Work-Energy Theorem

Kinetic Energy

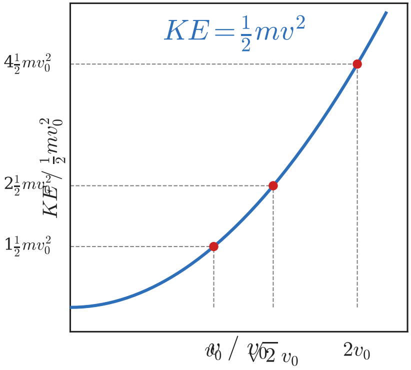
Work-Energy Theorem

Kinetic Energy

Kinetic energy is the energy an object possesses due to its motion:

$$KE = \frac{1}{2}mv^2$$

- Always ≥ 0 (scalar, never negative)
- Doubles when $v \to \sqrt{2}\,v$; quadruples when $v \to 2v$
- Same units as work: J



KE depends on v^2 — a car at 60 mph has *four times* the KE of the same car at 30 mph.

Outline of Part II: Kinetic Energy & the Work-Energy Theorem

Kinetic Energy

Work-Energy Theorem

The Work-Energy Theorem

Work-Energy Theorem

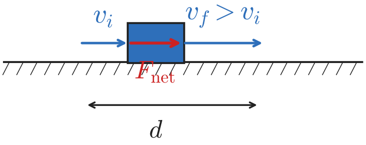
The *net* work done on an object equals the change in its kinetic energy:

$$W_{\text{net}} = \Delta KE = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2$$

This is Newton's second law in terms of energy. $F_{\text{net}} = ma$ and kinematics ($v_f^2 = v_i^2 + 2ad$) together give $W_{\text{net}} = F_{\text{net}}\, d = \Delta KE$.

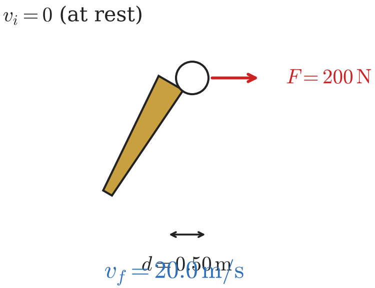
- $W_{\text{net}} > 0$: object speeds up
- $W_{\text{net}} < 0$: object slows down
- $W_{\text{net}} = 0$: constant speed

$$W_{\text{net}} = \Delta KE = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2$$



Work-Energy Theorem: Baseball Example

A 0.500 kg ball starts from rest. A bat exerts a net force of 200 N over a contact distance of 0.500 m. What is the ball's final speed?



$$W_{\text{net}} = F_{\text{net}}\, d = (200)(0.500) = 100\,\mathrm{J}$$

$$W_{\text{net}} = \tfrac{1}{2} m v_f^2 - 0$$

$$v_f = \sqrt{\frac{2\,W_{\text{net}}}{m}} = \sqrt{\frac{2(100)}{0.500}} = \boxed{20.0\,\mathrm{m/s}}$$

Check Your Understanding

A $1500\,\mathrm{kg}$ car traveling at $25.0\,\mathrm{m/s}$ brakes to a stop.
What is the net work done on the car by the braking force?

Come to lecture to see the answer.

Part III

Potential Energy & Conservation of Energy

Outline of Part III: Potential Energy & Conservation of Energy

Gravitational Potential Energy

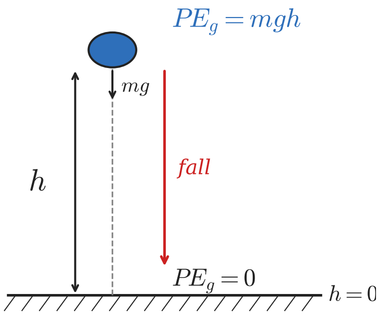
Gravitational PE is energy stored by position in a gravitational field (relative to a chosen reference height):

$$PE_g = mgh$$

- h measured from a reference point ($h = 0$)
- Only *changes* in PE_g matter physically
- Depends on mass, g, and height — not on path

📝 Mass Cancels

The speed gained falling a height h is the *same for all masses*: $v = \sqrt{2gh}$, since $PE_g = mgh$ and $KE = \frac{1}{2}mv^2$ both scale with m.



Outline of Part III: Potential Energy & Conservation of Energy

Conservative & Non-Conservative Forces

Conservative Forces

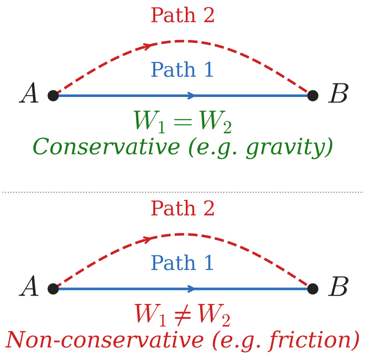
Work done is *path-independent*; energy can be stored and recovered.

- Gravity, spring (elastic) force
- Associated with a potential energy PE

Non-Conservative Forces

Work done *depends on path*; energy is permanently converted (often to heat).

- Friction, air drag, tension (when rope stretches)
- *Cannot* be associated with a PE



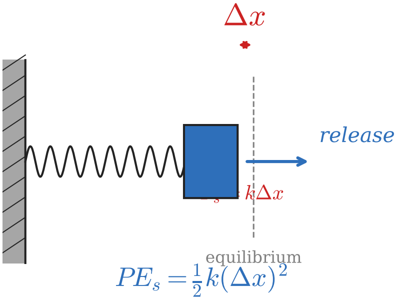
Outline of Part III: Potential Energy & Conservation of Energy

Elastic (Spring) Potential Energy

Energy stored in a compressed or stretched spring:

$$PE_s = \frac{1}{2}k(\Delta x)^2$$

- Δx = deformation from natural length
- Always ≥ 0 regardless of compression or extension
- Released as KE when spring returns to equilibrium



Recall Hooke's Law (Ch. 5)

$\vec{F}_s = k\Delta\vec{x}$ (restoring force of a spring)

Outline of Part III: Potential Energy & Conservation of Energy

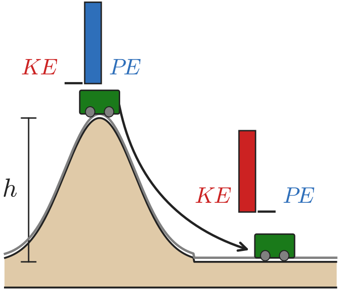
Conservation of Mechanical Energy

$$E_{\text{total}} = KE + PE$$

When only conservative forces act:

$$\Delta E_{\text{total}} = 0 \implies KE_i + PE_i = KE_f + PE_f$$

Pick the state where one term is simplest (e.g. $v = 0$ or $h = 0$), set it as your reference, then equate total energies at two states.



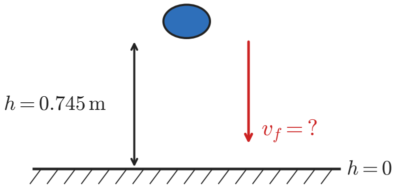
$$KE_i + PE_i = KE_f + PE_f$$

Conservation of Energy: Ball Drop

A ball is dropped from rest at $h = 74.5\,\text{cm}$. What is its speed just before it hits the ground?

$$mgh = \tfrac{1}{2}mv_f^2$$
$$v_f = \sqrt{2gh} = 3.82\,\text{m/s}$$

$$KE_i + PE_i = KE_f + PE_f$$
$$0 + mgh = \tfrac{1}{2}mv_f^2 + 0$$
$$v_f = \sqrt{2gh} = \sqrt{2(9.80)(0.745)}$$
$$\boxed{v_f = 3.82\,\text{m/s}}$$



📝 Note: mass cancels — result is independent of m.

Check Your Understanding

A roller-coaster car starts from rest at the top of a frictionless hill of height $h = 5.00\,\mathrm{m}$.

What is its speed at the bottom?

Come to lecture to see the answer.

Check Your Understanding

A spring with $k = 300\,\mathrm{N/m}$ is compressed $\Delta x = 0.200\,\mathrm{m}$ and launches a $0.120\,\mathrm{kg}$ block on a frictionless surface.

What is the block's speed after release?

Come to lecture to see the answer.

Part IV

Non-Conservative Forces & Energy Loss

Outline of Part IV: Non-Conservative Forces & Energy Loss

Work Done by Non-Conservative Forces

When NC Forces Are Present

General Energy Equation

Non-conservative forces change the total mechanical energy:

$$W_{\text{nc}} = \Delta E_{\text{total}} = \Delta KE + \Delta PE$$

Money Analogy

Think of E_{total} as your bank balance. Conservative forces move money between checking (KE) and savings (PE). Non-conservative forces (friction) are *taxes* — the money leaves the system permanently as heat.

- Friction: $W_{\text{nc}} = -f_k\, d < 0$
- Applied force: $W_{\text{nc}} = F_{\text{app}}\, d > 0$

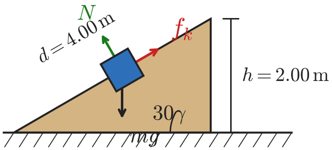
NC Forces: Ramp Example

A 3.00 kg block slides down a 4.00 m ramp inclined at $\theta = 30.0°$ from rest. Kinetic friction coefficient $\mu_k = 0.100$. Find the speed at the bottom.

$$h = (4.00)\sin 30.0° = 2.00\,\mathrm{m}$$

$$f_k = \mu_k mg\cos\theta = 0.100(3.00)(9.80)\cos 30.0° = 2.55\,\mathrm{N}$$

$$W_{\mathrm{nc}} = -f_k\, d = -(2.55)(4.00) = -10.2\,\mathrm{J}$$



$$W_{\mathrm{nc}} = \Delta KE + \Delta PE$$

$$-10.2 = \tfrac{1}{2}(3.00)v_f^2 - (3.00)(9.80)(2.00)$$

$$\boxed{v_f = 5.69\,\mathrm{m/s}}$$

Check Your Understanding

A 2.00 kg block starts from rest on a flat surface with $\mu_k = 0.300$. A horizontal force $F = 30.0\,\mathrm{N}$ pushes it 5.00 m. What is its final speed?

Come to lecture to see the answer.

Part V

Power & Efficiency

Outline of Part V: Power & Efficiency

Power

Efficiency

Power

Power is the rate at which work is done (or energy is transferred):

$$P = \frac{W}{t} = \frac{E}{t}$$

- SI unit: **watt** (W = J/s)
- 1 hp = 746 W (historical unit)
- Also: $P = Fv$ (when force is along velocity)

Device	Typical Power
LED light bulb	10 W
Hair dryer	1500 W
Car engine	150 kW
Large power plant	1 GW

Check Your Understanding

A $600\,\mathrm{W}$ electric motor lifts a $50.0\,\mathrm{kg}$ crate straight up for $30.0\,\mathrm{s}$ (assume 100% efficiency).

How high does it lift the crate?

Come to lecture to see the answer.

Outline of Part V: Power & Efficiency

Power

Efficiency

Efficiency

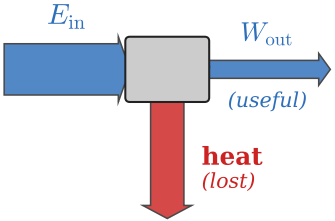
Definition

Efficiency measures how much of the input energy becomes useful output:

$$\eta = \frac{W_{\text{out}}}{E_{\text{in}}} \times 100\%$$

Always $\eta \leq 100\%$ (NC forces like friction always steal some energy).

$$\eta = \frac{W_{\text{out}}}{E_{\text{in}}} \times 100\%$$



Device/Activity	η
Human cycling	$\sim 25\%$
Car engine (gasoline)	$\sim 25\%$
LED bulb	$\sim 90\%$
Electric motor	$\sim 95\%$

Check Your Understanding

A machine receives $E_{\text{in}} = 5000\,\text{J}$ of energy and does $W_{\text{out}} = 1500\,\text{J}$ of useful work.

What is its efficiency?

Come to lecture to see the answer.

Next Lecture

Chapter 8: Linear Momentum and Collisions

- Momentum $\vec{p} = m\vec{v}$ and impulse
- Conservation of momentum
- Elastic and inelastic collisions
- Center of mass

Key connection: Just as the work-energy theorem links force to energy, the impulse-momentum theorem links force to momentum. Both are consequences of $F = ma$.